Thus the metric d_e induces a norm $\|\cdot\|_e$ on $E(M,Y)$ given by

$$\|f\|_e = d_e(f,0)$$

for all $f \in E(M,Y)$ whenever Y is a (real) normed linear space.

In particular, if Y is a Banach space, then $(E(M,Y), \|\cdot\|_e)$ is also a Banach space. □

Remark 3.3. In general, there is no specific metric defined on $Lip_0(M,N)$. However, being a subspace of $E(M,N)$, we can provide the restriction of d_e on it. But we do not know whether, in general, $Lip_0(M,N)$ is complete with this metric or not, even assuming the completeness of N.

Example 3.4. *In general, $E(M,N)$ and $C(M,N)$ are distinct spaces. For example, let $M = N = \mathbb{R}$ and consider the functions*

$$g(x) = \begin{cases} 2 & \text{if } x = 1 \\ x & \text{if } x \neq 1 \end{cases}$$

and $h(x) = x^2$. Then $g \in E(\mathbb{R},\mathbb{R}) \setminus C(\mathbb{R},\mathbb{R})$ and $h \in C(\mathbb{R},\mathbb{R}) \setminus E(\mathbb{R},\mathbb{R})$.

Now we consider the space $E_c(M,Y) = C(M,Y) \cap E(M,Y)$ as a normed subspace of $E(M,Y)$.

Proposition 3.5. *Consider $E_c(M,N) := C(M,N) \cap E(M,N)$ where M and N are metric spaces with distinguished points as 0. Then $E_c(M,N)$ is a closed subspace of $E(M,N)$ and hence is complete whenever N is so.*

Proof. Let (f_n) be a sequence in $E_c(M,N)$ such that it converges to $f \in E(M,N)$. We show that f is continuous.

Step I. Continuity of f at $0 \in M$.

Since $f_n \to f$, given $\epsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that $d_e(f_n, f) < \epsilon$ for all $n \geq n_0$. Thus $d(f_n(x), f(x)) \leq \epsilon d(x,0)$ for all $n \geq n_0$ and $x \in M$. Now f_{n_0} is continuous at 0 so there exists $\delta > 0$ such that $d(f_{n_0}(x), 0) < \frac{\epsilon}{2}$ whenever $d(x,0) < \delta$. (In fact, $f_{n_0}(0) = 0$.) Consider $\delta' = min\{\delta, \frac{1}{2}\} > 0$, then for $d(x,0) < \delta'$, we have

$$d(f(x), f(0)) = d(f(x), 0) \leq d(f_{n_0}(x), f(x)) + d(f_{n_0}(x), 0)$$